Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller

Verification Plan

Reference	TBD **[TBD-1]**
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1.0 Purpose and Scope

This plan states how each of the 140 software requirements [3, Secs. 4.0–7.0] is verified: by which method, in which activity, and against which success criterion. It also runs every ConOps scenario from start to end, to show that the System behaves as the ConOps describes [4, Sec. 6.0], and checks that the chosen hardware provides what the System needs. It is the verification plan for the software and for the hardware it needs, following the outline of a verification and validation plan [1, App. I, pp. 216–222]; for the software, it is the test plan [2, SWE-065, Sec. 4.5.2, p. 35].

2.0 Documents

[1] NASA, *NASA Systems Engineering Handbook*, NASA/SP-2016-6105 Rev 2, 2016.
[2] NASA, *NASA Software Engineering Requirements*, NPR 7150.2D, 2022.
[3] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Software Requirements Specification*, Issue 1, Revision 0, Draft.
[4] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Concept of Operations*, Issue 1, Revision 1, Draft.
[5] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Software Architecture*, Issue 1, Revision 0, Draft.
[6] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Interface Design Description*, Issue 1, Revision 0, Draft.
[7] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Hazard Analysis*, Issue 1, Revision 1, Draft.

3.0 System Description

The System has three software items: the ground software, the onboard computer software and the controller firmware [5, Sec. 3.0]. On the bench, test tools surround them to cause faults on purpose and to record what happens (Figure 1). The test tools exist only on the bench [3, REQ-101].

Radio simulator Software that takes the place of the radio equipment, which lies outside the System [4, Sec. 1.2.2]. It carries envelopes between the ground software and

	the onboard computer, opens and closes contact windows, and mimics losses, noise and attacks: it drops, repeats, changes or forges messages.
Link tool	On each data link. Cuts and restores it; changes bytes on it.
Power switch	Cuts the controller's power at a chosen moment.
Debug probe	Reads and changes the controller's memory, and stops its core, without its software.
Test firmware	Controller firmware built to fail on purpose: it hangs, restarts over and over, or is never judged good.
Test script	Runs each activity, stops programs on the onboard computer, restarts it, and time-stamps every event from one clock.

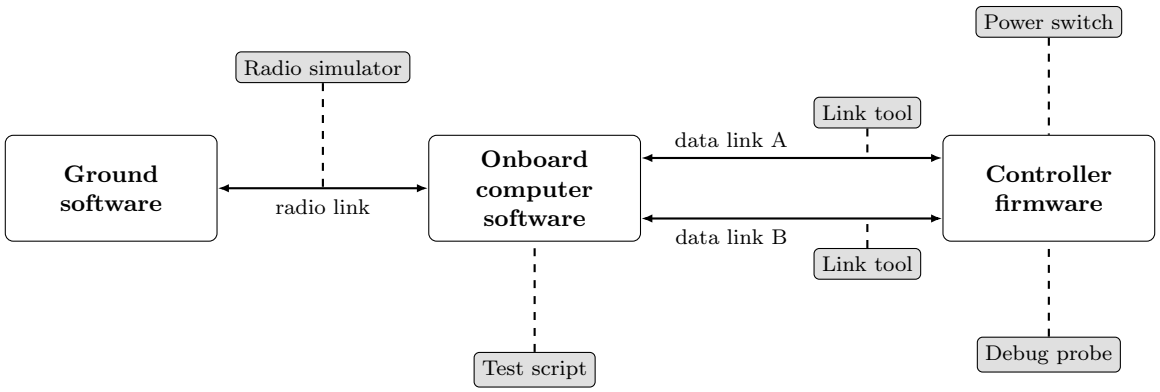


Figure 1: The bench. Grey: test tools; dashed: where each one acts. The reprogramming links are not drawn; the debug probe shares the controller's debug port with reprogramming link A, one at a time.

4.0 Verification Methods

Each requirement is verified by the method in the "Verified by" column of the requirements specification [3, Secs. 4.0–7.0]. The four methods [1, p. 93]:

Test	"The use of an end product to obtain detailed data needed to verify performance or provide sufficient information to verify performance through further analysis."
Analysis	"The use of mathematical modeling and analytical techniques to predict the suitability of a design to stakeholder expectations based on calculated data or data derived from lower system structure end product verifications."
Inspection	"The visual examination of a realized end product." It "can include inspection of drawings, documents, or other records."
Demonstration	"Showing that the use of an end product achieves the individual specified requirement."

Tests and demonstrations run at one of four levels:

COMPONENT	One component of the architecture [5, Sec. 6.0] alone; test code stands in for its neighbours.
ITEM	One software item alone; test tools stand in for its neighbours.
INTEGRATION	Two or more items together, over their real interfaces.
SYSTEM	All three items together, with faults caused on purpose.

5.0 Test Articles

The test articles are breadboards: "a low fidelity unit that demonstrates function only", which "often uses commercial and/or ad hoc components and is not intended to provide definitive

information regarding operational performance” [1, p. 124]. Besides the hardware needs of the architecture [5, Sec. 8.0], the bench must let the test tools do the following:

1. Cut and restore each data link on command, and change bytes on it.
2. Cut the controller's power at a chosen moment, including during a write to its memory.
3. Restart the onboard computer, and stop any of its programs.
4. Read and change the controller's memory without its software, sharing the debug port with the onboard computer one at a time.
5. Time-stamp events on every unit from one clock.
6. Open and close contact windows, and drop, repeat, change or forge messages on the radio link.

The hardware is chosen in a trade study against these needs and the hardware needs of the architecture, and is named here **[TBD-3]**.

6.0 Component and Item Verification

Each component, then each software item, is tested alone. A test passes when its success criterion holds in every run. Each activity is broken into test procedures, one per condition, once the test articles are chosen [2, SWE-065, Sec. 4.5.2, p. 35]. The activities of the three software items run at the ITEM level.

6.1 Components

Every component of the architecture is tested alone, at the COMPONENT level [2, SWE-062, Sec. 4.4.5, p. 34]. Its tests are written with its code, and its results are repeatable [2, SWE-186, Sec. 4.4.6, p. 35]. The safety-critical components reach 100 % modified condition/decision coverage [3, REQ-108]. The test cases are kept with the code, not listed in this plan.

6.2 Controller Firmware

6.3 Controller Firmware

Activity	Success criterion	Requirements
Test-1 Modes	Every transition of the ConOps mode table is driven. The controller reports exactly one mode at all times, accepts each command only in the modes that allow it, and makes no other transition.	REQ-001 to 003
Test-2 Safe state	After every start, and after every function that ends, the outputs are in the safe state before any command acts. A safe-state command reaches the safe state from every mode.	REQ-004, REQ-102, REQ-106
Test-3 Image checks	An image with one bit changed, or with either signature invalid, is not run, and the current copy keeps running.	REQ-005, REQ-006
Test-4 Trial	New firmware runs a trial of the set length, runs its trial tests and reports the result, and is confirmed only on the operator's CONFIRM after the tests passed. It is given up, and the previous copy runs, at once when a trial test fails, when not confirmed in time, when it hangs, or on the operator's order. The switch happens at the set moment.	REQ-011 to 014, REQ-020, REQ-021, REQ-113, REQ-139, REQ-140
Test-5 Failed starts	After the set number of failed starts, power cuts during a trial included, the previous copy runs. The count survives restarts and power cuts. Failed firmware is not started again unless ordered.	REQ-015 to 017, REQ-019
Test-6 Fall-back	An update never changes the previous copy or the last-resort firmware, checked by reading memory back. When the previous copy also fails, the last-resort firmware runs, receives new firmware, and stays in FALLBACK until the operator acts.	REQ-007, REQ-008, REQ-010, REQ-018, REQ-117
Test-7 Restart	Firmware made to hang restarts on its own within the restart time and resumes operation.	REQ-022, REQ-023

Activity	Success criterion	Requirements
Test-8 Independence	The controller keeps working while the onboard computer restarts, with both data links cut, and through a run of [**TBD-2**] without commands.	REQ-024 to 026
Test-9 Command checks	Each command not allowed in the mode, with impossible values, not armed, armed too long ago, past its expiry, or scheduled while the clocks disagree is refused with its reason. A command sent again runs once and is answered ALREADY DONE. A held command changed before its run time is refused. No single changed bit of the stored arm state arms it.	REQ-027 to 032, REQ-133, REQ-134, REQ-137
Test-10 Answers	Every command gets RECEIVED, then DONE only after its effect is seen, or REFUSED with its reason.	REQ-033 to 035
Test-11 Reports and faults	Health reports arrive at the set rate. Each injected fault, failed self-test and fall-back produces its event report. A changed bit in the controller's memory is detected and leads to the safe state. Traffic from the controller never stops the onboard computer reaching it.	REQ-037 to 042
Test-12 Clock	With the onboard computer's clock stopped or wrong, the controller's clock keeps time from its own source and changes only on SET CONTROLLER TIME. Every report carries both times.	REQ-131, REQ-132, REQ-135

6.4 Onboard Computer Software

Activity	Success criterion	Requirements
Test-13 Image checks	Images not released by the operator, damaged, older than the minimum version, or built for another controller are refused with their reason. An image changed after its last transfer is refused before delivery.	REQ-043 to 047
Test-14 Upload and delivery	An upload cut at random chunks resumes where it stopped. Delivery runs while the controller works. Delivery during a trial is refused with its reason. A cancelled update leaves the working firmware running and frees the storage.	REQ-048 to 052
Test-15 Limits	SET LIMITS changes the minimum version and the controller identity, and only after an arming command.	REQ-053
Test-16 Command intake	Frames not encrypted with the right key, with either signature invalid, sent again, or past their expiry are refused and logged. Commands that do not arrive over the radio link are refused. A dangerous command without a valid arming command is not passed on.	REQ-064 to 067, REQ-112, REQ-127, REQ-129
Test-17 Scheduled commands	A scheduled command is passed on at its run time, once, also across a restart and a power cut. A held command with a changed bit is not passed on.	REQ-068, REQ-069, REQ-136
Test-18 Report store	Every report is numbered, time-tagged, checked on arrival, kept through power cuts and restarts, and sent at the next contact, events first. When storage fills, the oldest health reports go first, and the number and time span dropped are reported. A clock difference above the limit is reported, and a correction of the onboard clock that does not rely on it is accepted.	REQ-070 to 079
Test-19 Data links	With either data link cut, all traffic continues on the other. Each loss and each recovery is reported. With both cut, the reprogramming link still works.	REQ-080 to 083
Test-20 Key replacement	A new key is accepted only in a command proven with the current one. Afterwards only the new key works. A replacement cut at any step leaves the current key working.	REQ-084 to 086
Test-21 Command log	The ground receives a log of every command acted on, including one injected past the ground.	REQ-087
Test-22 Recovery	A silent controller is noticed and brought back, and the attempt is recorded. After a program is killed, and after the whole onboard computer restarts, the update in progress resumes.	REQ-061 to 063

6.5 Ground Software

Activity	Success criterion	Requirements
Test-23 Ground checks	Every command and release leaves signed twice. Malformed commands, and releases for another controller, older than the minimum version or not the tested build, are not sent. Each signed image passes the onboard computer's checks before it is sent. A planned sequence stops at the first refusal or missing answer.	REQ-088 to 090, REQ-114 to 116
Test-24 Ground display	A damaged report is marked on arrival. Every command still waiting for its answers, and the running firmware, are shown.	REQ-093 to 095
Demonstration-1 Operator access	Someone without the operator's credentials cannot use the operator interface.	REQ-091
Demonstration-2 Planning	The operator plans commands, uploads and report retrieval for coming contacts.	REQ-097

7.0 System Verification

The items are then verified together. The restore and the interfaces are verified at the INTEGRATION level; fault injection and scenario runs at the SYSTEM level.

7.1 Restore

Activity	Success criterion	Requirements
Test-25 Restore	With the controller's memory erased, and with its bootloader damaged, a restore through each reprogramming link writes only checked images, puts the last-resort firmware first and clears the kept data. On order, it puts back the bootloader and the main firmware. The onboard computer restarts the controller, checks its copy and the links between uses, and tells the ground the result.	REQ-054, REQ-057 to 060, REQ-118, REQ-120 to 122, REQ-124
Demonstration-3 Restore on order	The operator orders a restore while firmware runs, and it completes.	REQ-055

7.2 Interfaces

Activity	Success criterion	Requirements
Test-26 Interfaces	Every message on each interface matches the Interface Design byte for byte. A frame with any byte changed is dropped on every link.	REQ-107

7.3 Fault Injection

Activity	Success criterion	Requirements
Test-27 Fault injection	Each fault of Table 1 is injected during operation. Every detected error is handled, the response matches the ConOps scenario, and no hazard occurs.	REQ-105

Fault	Caused by	ConOps scenario
Radio link cut during an upload	RADIO SIMULATOR	Scenario-200
Long gap between contacts fills the report storage	RADIO SIMULATOR	Scenario-210
Forged command, or old command sent again	RADIO SIMULATOR	Scenario-300
Firmware damaged in transfer	RADIO SIMULATOR	Scenario-400
New firmware hangs, restarts over and over, or is never judged good	TEST FIRMWARE	Scenario-430 to 450
Previous firmware also unusable	DEBUG PROBE	Scenario-460
No firmware can run	DEBUG PROBE	Scenario-480
One data link cut, then restored; both cut	LINK TOOL	Scenario-500 to 510
Controller goes silent	DEBUG PROBE	Scenario-520
Program on the onboard computer stopped	TEST SCRIPT	Scenario-530
Whole onboard computer restarted	TEST SCRIPT	Scenario-535
Power cut during a firmware write, and during a trial	POWER SWITCH	Scenario-600, 605
Bit changed in stored firmware	DEBUG PROBE	Scenario-610
Bit changed in the memory the controller runs from	DEBUG PROBE	Scenario-620

Table 1: Faults caused on purpose [4, Sec. 6.2].

7.4 Scenario Runs

Activity	Success criterion	Requirements
Test-28 Scenario runs	Every ConOps scenario, nominal and off-nominal, is run from start to end, and each shows the outcome the ConOps states.	—

7.5 Inspections and Analyses

Activity	Success criterion	Requirements
Inspection-1 Separate code	The link maps and source lists of the parts each requirement keeps apart share no file, library or symbol.	REQ-009, REQ-099, REQ-119, REQ-123, REQ-126, REQ-130
Inspection-2 Records	The test logs hold every command received and answer sent. The ground archive holds every report sent. The recoverable copy of the keys restores them.	REQ-036, REQ-092, REQ-096
Inspection-3 Shared list	The ground and the onboard software take their commands and refusal reasons from one source file.	REQ-100
Inspection-4 Operating builds	The operating builds contain no fault-injection function.	REQ-101
Inspection-5 Code rules	Tool reports show a complexity of 15 or lower for each safety-critical component, conformance to the coding standard, and no open static-analysis finding.	REQ-109 to 111
Inspection-6 Ways back in	No controller firmware writes the option bytes or the boot settings, and the factory loader driver never sends Readout Protect or Write Protect.	REQ-125
Inspection-7 Keys and links	The two signing keys and the encryption key are kept in three separate places. The two data links are of different kinds.	REQ-128, REQ-138
Analysis-1 Fault tolerance	The hazard analysis, run again on the built System, finds at least three failures for each catastrophic hazard and two for each critical one, the accepted exceptions aside, and no hazard started by a single software event.	REQ-098, REQ-103
Analysis-2 Response times	For each off-nominal condition, the response time measured in Test-27 is shorter than the time to the hazard.	REQ-104

Activity	Success criterion	Requirements
Analysis-3 Restore independence	The restore sequences of the Interface Design use no controller software: the core is held stopped, or the factory loader runs from read-only memory.	REQ-056
Analysis-4 Coverage	The coverage tool reports 100 % modified condition/decision coverage of the safety-critical software over all tests.	REQ-108

7.6 Hardware

These two activities check the hardware itself, against the hardware needs of the architecture [5, Sec. 8.0] and of Section 5.0. They run once the hardware is chosen.

Activity	Success criterion	Needs
Inspection-8 Hardware documents	Every hardware need of the architecture, and every need of Section 5.0, is shown met by the chosen hardware's datasheets and manuals, each with its page.	Architecture Section 8.0; Section 5.0
Test-29 Hardware measurements	The needs that documents cannot settle are measured on the test articles: two signatures are checked at start-up within the restart time of REQ-023; software cannot stop the watchdog; the clock runs from its own source, at a measured rate; the debug port and the factory loader reach all of memory with the controller held in reset.	Architecture Section 8.0

8.0 Limits of Bench Verification

The System runs on commercial development boards: "It can show how the System behaves, not that it would survive launch or space" [4, Sec. 1.1.2]. The requirements below are verified only in part. What the bench cannot show is the risk left, accepted by the author.

Requirement	The bench shows	The bench cannot show
REQ-041	A bit changed on purpose is detected	Real radiation: its rate, several bits changed at once, damage to the chip
REQ-026	Correct operation through the run of Test-8	Operation for longer than that run
REQ-064, REQ-129	Forged and changed frames are refused	Resistance to every possible forgery, which rests on Ed25519 and AES-256 in Galois/Counter Mode
REQ-098, REQ-103	Nothing; Analysis-1 shows it	That the analysis found every combination of failures
REQ-080, REQ-138	Failover between two data links over bench wires; the second is a UART	The CAN link; two links of different kinds; the length and electrical noise of a spacecraft harness
REQ-016, REQ-077	A reset at a chosen moment, in place of a power cut	A true loss of power
REQ-123	A restore through SWD driven by the ST-LINK	SWD driven by the onboard computer's own driver and pins
REQ-067	Contact windows and losses made by the radio simulator	The real radio equipment, which lies outside the System
All	Behaviour at room temperature and pressure	Temperature, vacuum, vibration and launch

A Verification Matrix

Each requirement, its verification method [3, Secs. 4.0–7.0], and the activities that verify it [1, App. D, p. 201].

Req.	Method	Activities	Req.	Method	Activities
REQ-001	TEST	Test-1	REQ-071	TEST	Test-18
REQ-002	TEST	Test-1	REQ-072	TEST	Test-18
REQ-003	TEST	Test-1	REQ-073	TEST	Test-18
REQ-004	TEST	Test-2	REQ-074	TEST	Test-18
REQ-005	TEST	Test-3	REQ-075	TEST	Test-18
REQ-006	TEST	Test-3	REQ-076	TEST	Test-18
REQ-007	TEST	Test-6	REQ-077	TEST	Test-18
REQ-008	TEST	Test-6	REQ-078	TEST	Test-18
REQ-009	INSPECTION	Inspection-1	REQ-079	TEST	Test-18
REQ-010	TEST	Test-6	REQ-080	TEST	Test-19
REQ-011	TEST	Test-4	REQ-081	TEST	Test-19
REQ-012	TEST	Test-4	REQ-082	TEST	Test-19
REQ-013	TEST	Test-4	REQ-083	TEST	Test-19
REQ-014	TEST	Test-4	REQ-084	TEST	Test-20
REQ-015	TEST	Test-5	REQ-085	TEST	Test-20
REQ-016	TEST	Test-5	REQ-086	TEST	Test-20
REQ-017	TEST	Test-5	REQ-087	TEST	Test-21
REQ-018	TEST	Test-6	REQ-088	TEST	Test-23
REQ-019	TEST	Test-5	REQ-089	TEST	Test-23
REQ-020	TEST	Test-4	REQ-090	TEST	Test-23
REQ-021	TEST	Test-4	REQ-091	DEMONSTRATION	Demonstration-1
REQ-022	TEST	Test-7	REQ-092	INSPECTION	Inspection-2
REQ-023	TEST	Test-7	REQ-093	TEST	Test-24
REQ-024	TEST	Test-8	REQ-094	TEST	Test-24
REQ-025	TEST	Test-8	REQ-095	TEST	Test-24
REQ-026	TEST	Test-8	REQ-096	INSPECTION	Inspection-2
REQ-027	TEST	Test-9	REQ-097	DEMONSTRATION	Demonstration-2
REQ-028	TEST	Test-9	REQ-098	ANALYSIS	Analysis-1
REQ-029	TEST	Test-9	REQ-099	INSPECTION	Inspection-1
REQ-030	TEST	Test-9	REQ-100	INSPECTION	Inspection-3
REQ-031	TEST	Test-9	REQ-101	INSPECTION	Inspection-4
REQ-032	TEST	Test-9	REQ-102	TEST	Test-2
REQ-033	TEST	Test-10	REQ-103	ANALYSIS	Analysis-1
REQ-034	TEST	Test-10	REQ-104	ANALYSIS	Analysis-2
REQ-035	TEST	Test-10	REQ-105	TEST	Test-27
REQ-036	INSPECTION	Inspection-2	REQ-106	TEST	Test-2
REQ-037	TEST	Test-11	REQ-107	TEST	Test-26
REQ-038	TEST	Test-11	REQ-108	ANALYSIS	Analysis-4
REQ-039	TEST	Test-11	REQ-109	INSPECTION	Inspection-5
REQ-040	TEST	Test-11	REQ-110	INSPECTION	Inspection-5
REQ-041	TEST	Test-11	REQ-111	INSPECTION	Inspection-5
REQ-042	TEST	Test-11	REQ-112	TEST	Test-16
REQ-043	TEST	Test-13	REQ-113	TEST	Test-4
REQ-044	TEST	Test-13	REQ-114	TEST	Test-23
REQ-045	TEST	Test-13	REQ-115	TEST	Test-23
REQ-046	TEST	Test-13	REQ-116	TEST	Test-23
REQ-047	TEST	Test-13	REQ-117	TEST	Test-6
REQ-048	TEST	Test-14	REQ-118	TEST	Test-25
REQ-049	TEST	Test-14	REQ-119	INSPECTION	Inspection-1
REQ-050	TEST	Test-14	REQ-120	TEST	Test-25
REQ-051	TEST	Test-14	REQ-121	TEST	Test-25
REQ-052	TEST	Test-14	REQ-122	TEST	Test-25
REQ-053	TEST	Test-15	REQ-123	INSPECTION	Inspection-1
REQ-054	TEST	Test-25	REQ-124	TEST	Test-25
REQ-055	DEMONSTRATION	Demonstration-3	REQ-125	INSPECTION	Inspection-6
REQ-056	ANALYSIS	Analysis-3	REQ-126	INSPECTION	Inspection-1
REQ-057	TEST	Test-25	REQ-127	TEST	Test-16
REQ-058	TEST	Test-25	REQ-128	INSPECTION	Inspection-7
REQ-059	TEST	Test-25	REQ-129	TEST	Test-16
REQ-060	TEST	Test-25	REQ-130	INSPECTION	Inspection-1
REQ-061	TEST	Test-22	REQ-131	TEST	Test-12
REQ-062	TEST	Test-22	REQ-132	TEST	Test-12
REQ-063	TEST	Test-22	REQ-133	TEST	Test-9
REQ-064	TEST	Test-16	REQ-134	TEST	Test-9
REQ-065	TEST	Test-16	REQ-135	TEST	Test-12
REQ-066	TEST	Test-16	REQ-136	TEST	Test-17
REQ-067	TEST	Test-16	REQ-137	TEST	Test-9
REQ-068	TEST	Test-17	REQ-138	INSPECTION	Inspection-7
REQ-069	TEST	Test-17	REQ-139	TEST	Test-4
REQ-070	TEST	Test-18	REQ-140	TEST	Test-4

B Acronyms

ConOps Concept of Operations
TBD To Be Determined
TBR To Be Resolved

C Open Items

The values below are not yet known.

TBD-1 Project name and document reference.
TBD-2 Length of the run without commands in Test-8.
TBD-3 The test articles, chosen in a hardware trade study (Section 5.0).

All statements in this document were stated or accepted by the author on 12 October 2026.